Problem-Solution Fit

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
This canvas explains how the OptiCrop: Smart Agricultural Production Optimization Engine addresses farmers' crop selection problems by using machine learning to recommend the best crop based on soil parameters and agricultural data..

<table>
<tr>
<td>1. CUSTOMER SEGMENT(S) [CS]
• Small and Medium Scale Farmers</td>
<td>6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]
Limited budget.</td>
<td>5. AVAILABLE SOLUTIONS
PROS & CONS [AS]
Accurate but expensive and not always available.</td>
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<td>2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]
• Difficulty in selecting the best crop for soil conditions.
• Low crop yield due to wrong crop selection</td>
<td>9. PROBLEM ROOT / CAUSE [RC]
• Lack of scientific crop recommendation systems.
• Insufficient knowledge of soil parameters.</td>
<td>7. BEHAVIOR
+ ITS INTENSITY [BE]
• Farmers usually depend on traditional methods and experience.</td>
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<td>3. TRIGGERS TO ACT [TR]
• Decreasing crop production.
• Soil nutrient imbalance.</td>
<td rowspan="2">10. YOUR SOLUTION [SL]
OptiCrop: Smart Agricultural Production Optimization Engine uses Machine Learning algorithms to analyze soil parameters and recommend the most suitable crop, helping farmers improve productivity and reduce losses.</td>
<td>8. CHANNELS OF BEHAVIOR [CH]
ONLINE:Mobile applications, websites, social media, agricultural portals.</td>
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<td>4. EMOTIONS
BEFORE / AFTER [EM]
Before: uncertain about crop selection.
After:optimistic about farming decisions.</td>
<td>OFFLINE
Agricultural officers, local markets, farmer meetings, village communities.</td>
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